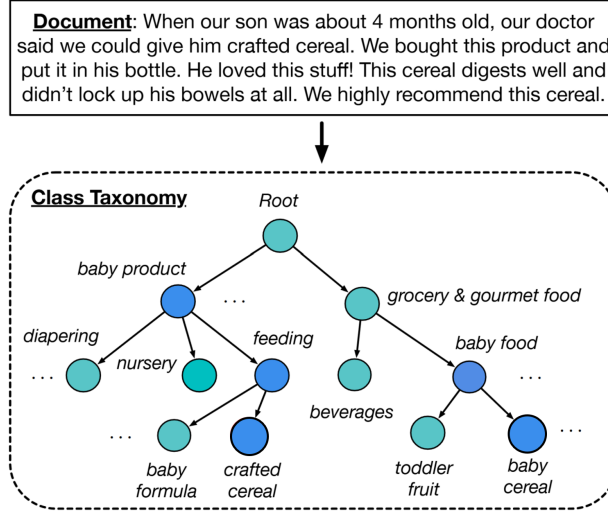


Task Description

Your task is to perform product review classification without using any labeled data. This problem is called a **Hierarchical Multi-Label Text Classification**, where each review can be associated with one or more product categories that are organized in a hierarchical taxonomy. You are encouraged to use [this paper](#) as a key reference for this project.



An example of hierarchical product taxonomy used in the task.

Task Definition. Formally, the task input includes an unlabeled text corpus $\mathcal{D}$ and a directed acyclic graph $\mathcal{T} = (\mathcal{C}, \mathcal{R})$ that defines the class taxonomy. Each $c_i \in \mathcal{C}$ represents a product class in the taxonomy. Each edge $\langle c_i, c_j \rangle \in \mathcal{R}$ indicates a parent-child relation, where class c_j is a sub-class of c_i . For example, one such edge may exist between $s_i = \text{“baby food”}$ and $s_j = \text{“baby cereal”}$.

The goal is to learn a multi-label text classifier $f(\cdot)$ that maps a document d into a binary label vector $\mathbf{y} = [y_1, \dots, y_{|\mathcal{C}|}]$, where $y_i = 1$ if d belongs to class c_i , and $y_i = 0$ otherwise. In the given dataset, each document is associated with at least two and at most three labels.

Task Setup. You are provided with the following resources:

1. **Training corpus:** 29,487 product reviews without class labels.
2. **Classes:** 531 product categories.
3. **Class hierarchy:** A taxonomy file that defines parent-child relationships among classes (each line represents one relation).
4. **Class-related keywords:** A list of keywords associated with each product class.
5. **Test corpus:** 19,658 product reviews for evaluation.

You may use **all** of the above information and up to **1,000 LLM calls** to complete the task. That is, you are allowed to use the test corpus information during training.